Fiscal Year:	FY 2017	**Task Last Updated:**	FY 04/07/2017
PI Name:	Gibble, Kurt Ph.D.		
Project Title:	Contributions to the Evaluation of the ACES Clock PHARAO		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802-6300	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/08/2012	**End Date:**	12/31/2018
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2018 per PI (Ed., 3/10/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	ACES (Atomic Clock Ensemble in Space) is a European Space Agency (ESA) project that will test fundamental physics, search for the time variation of fundamental constants, and enable highly precise frequency comparisons between clocks on Earth. It is planned to operate for at least 18 months on the International Space Station. PHARAO (Projet d'Horloge Atomique par Refroidissement d'Atomes en Orbit) is the laser-cooled, microgravity, atomic cesium clock of ACES that provides state-of-the-art accuracy, with a goal of 1 part in 10^(16) accuracy. We will assist in the accuracy evaluation of PHARAO, a central scientific objective of the ESA mission. We will contribute to the analysis of the three largest sources of systematic uncertainty: clock frequency shifts due to the collisions of ultracold atoms, a Doppler shift known as the distributed cavity phase shift, and a frequency shift due to the lensing of the atomic wavepackets by the clock's microwave field. The current working uncertainty budget projects 2.2×10^(-16) accuracy. The proposed work will		

	reduce the largest contribution to the clock's uncertainty to negligible levels and investigate significant reductions of the others, with a goal of reaching 10^(-16) accuracy. The work will help define the limits of PHARAO's accuracy and identify workable procedures to demonstrate its accuracy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work on microwave cavities produced an improved design that will be used for Cs and Rb fountain clocks. These are being incorporated into new clocks being built at the British, South Korean, Polish, Canadian, and Indian National Laboratories. The new cavity design allows significantly reduced clock errors due to Doppler shifts. Our work has also led to a deeper understanding of the microwave lensing frequency shift that is being used as a correction of International Atomic Time and, more broadly, our work contributes to scientific progress towards anticipated future optical frequency atomic clocks.
Task Progress:	The primary focus of our work on the accuracy evaluation of PHARAO was to evaluate Doppler shifts known as cavity phase shifts. These include end-to-end and distributed cavity phase shifts. Using our previous simulations of clock shifts due to cavity phase, from both the known wall losses as well as potential inhomogeneities of the wall losses, we analyzed their dependence on launch velocity and optimized the operation procedures while on orbit to minimize cavity phase. Our approach is a new solution for the accuracy evaluation of these errors, and is expected to allow PHARAO to achieve its accuracy goals. A paper with our French collaborators on the qualifications of the PHARAO clock is in preparation. Related to the cold collision frequency shift of PHARAO, we have continued analysis of our precise measurements of Cesium scattering phase shifts in an atomic fountain clock. We extensively analyzed the thermal averages and the effects of clipping in our fountain. We developed an analytic model and showed that it reproduced known results and agrees with our Monte-Carlo model of the fountain, when no apertures are present in either. These are convincing tests of our Monte-Carlo model. The effects of the apertures are small and easily verifiable and we refined our treatment of these in our Monte-Carlo simulation, which gave small improvements to the energy distributions. We are investigating if measurements of the collision shift in PHARAO may be able to clarify the theoretical model. Along these lines, we analyzed collision shifts in a fountain. We developed both analytic and a full Monte-Carlo simulation of the fountain, which gave a preliminary shift close to the theoretical value. This gives significant confidence for the predictions of the PHARAO collision shift. Work continues on this analysis and to understand whether measured collision shifts in microgravity may clarify the theoretical model of Cs-Cs scattering. Our broader effort on future microgravity optical-frequency atomic clocks focused on candidates with small sensitivities to stray electric fields. Our primary candidate is an optical lattice clock based on neutral cadmium. This clock requires blue and ultraviolet laser sources. We investigated several approaches and successfully generated sufficient ultraviolet power.
Bibliography Type:	Description: (Last Updated: 05/28/2026)
Articles in Peer-reviewed Journals	Szymaniec K, Lea SN, Gibble K, Park SE, Liu K, Glowacki P. "NPL Cs fountain frequency standards and the quest for the ultimate accuracy." Journal of Physics: Conference Series. 2016;723:012003. (8th Symposium on Frequency Standards and Metrology 2015.) https://doi.org/10.1088/1742-6596/723/1/012003 , Jul-2016